

# Keep every promise, grow the rest

Investment strategy and research report for Laura Gao  
Wharton Global High School Investment Competition 2026-27

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Team Caplet · September 2026

Prepared for Team Caplet with an AI research assistant (Claude Code). It is research and inspiration, not text to submit: the team decides the strategy, writes its Trading Notes and IPS in its own words, and records AI use as Wharton requires. "Model" figures are computer estimates, not forecasts. Every trade is subject to the team's vote and a check of the trading platform.

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# Executive summary

Laura Gao has asked us to invest \$450,000 so that she can fund her creative residency in Taiwan. She needs ten payments of \$50,000 a year from 2033, made "with a high degree of certainty", and she wants to contribute responsibly to the residency's building while keeping some flexibility.

**Our recommendation: keep every promise, grow the rest**

- 1. **January 2027.** Her first \$300,000 buys ten U.S. government bonds, one maturing just before each \$50,000 payment. From that day, the payments no longer depend on markets.
- 2. **January 2028.** Her second deposit of \$150,000 buys a bond that repays \$150,000 before 2033. This is money for the building that she owns outright. The remaining ~\$40,000 goes into one global stock fund.
- 3. **2031 and 2033.** She tells partners a range: the \$150,000 she owns, up to that plus half of the stock fund. In 2033 the bonds pay the residency and the building gift is made. No trades are needed after 2028.

|                                  |                                                            |                                                                |                                                               |
|----------------------------------|------------------------------------------------------------|----------------------------------------------------------------|---------------------------------------------------------------|
| 10 of 10                         | \$150k                                                     | \$207k                                                         | \$169k                                                        |
| payments secured in January 2027 | building money owned three years before partners are asked | typical money for the building and flexibility in 2033 (model) | the same figure in the worst six years of U.S. market history |

**Why this is the right plan for Laura**

- **Her promises do not depend on markets.** Plans that rely on markets miss a payment up to 46% of the time if her second deposit never arrives (model). Ours does not, once the bonds are bought.
- **She never has to take a number back.** The bottom of her 2031 range is money she already owns, and the gift can never exceed the top she announces.
- **Bad years hurt less.** In the worst stretch of market history, our plan still leaves \$169,000 for the building, against \$120,000 for the next-best design.
- **It is simple.** Two purchases on two dates. Nothing to manage in 2031.

**What it costs her**

Less exposure to the stock market (about 9% of her money from 2028 to 2032) and a smaller share of a boom. For a founder whose public name rides on these promises, we believe this is the right trade, and we say so openly.

**How we reached it**

We ran a research program in eight phases: we logged every official rule, asked 472 questions from ten angles, kept the 48 that could change a decision, had twelve specialists answer them with auditors checking the work, asked four teams to attack the plan, settled one final design, tested it

blind against five other common approaches, and checked every number one last time. In the blind test, all three reviewers playing Laura ranked our plan first.

#### **What we discovered along the way**

- This year's trading rules are public, and the \$300,000 is only just enough to buy the ten payments. On most days of 2026 it would not have been.
- Two official competition documents were missing from our files; they are now filed.
- The famous quote in the case cannot be traced to Laura, and a popular story about her comes from a reporter. Neither is used.
- The real field is about 2,300 finishing teams, not 6,300.

#### **What we need from the team**

Three recorded votes before the first trade, a check of the trading platform's limits, and trades placed by 2 October. The Trading Notes are due 23 October and the Investment Policy Statement on 6 November.

# Understanding Laura

Laura is a bestselling graphic novelist, illustrator, entrepreneur and educator. Her income comes from publishing advances, speaking and licensing. She plans a creative residency in Taiwan and wants to fund it reliably.

| What she asked for                                                                               | What it means for her portfolio                                               |
|--------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------|
| Ten payments of \$50,000, 2033-2042, with "a high degree of certainty", from the portfolio alone | These come first. Secure them with bonds rather than hope markets cover them. |
| A responsible contribution to the building                                                       | Own a floor early, then add a share of the growth that actually happens.      |
| A credible range for partners in 2031                                                            | The bottom of the range must be money she already has.                        |
| Flexibility as the project develops                                                              | Some money stays hers after 2033.                                             |
| Balance growth with protecting capital                                                           | Take market risk only with money nobody depends on.                           |

## The insight that shaped everything

Laura's career is already her biggest risk. A weak year for her books would also shrink her second deposit. So her portfolio should not add the same kind of risk to the money her promises depend on. We protect first and grow second.

## Her money, in and out

- **In:** \$300,000 in January 2027 and \$150,000 in January 2028. Nothing else before 2033.
- **Out:** \$50,000 every January from 2033 to 2042, plus the building contribution in 2033.
- Her living costs, taxes and Taiwan's legal questions are handled elsewhere.

# The plan, year by year



Only the first two dates need trades. After that the plan runs itself.

## Where her \$450,000 goes

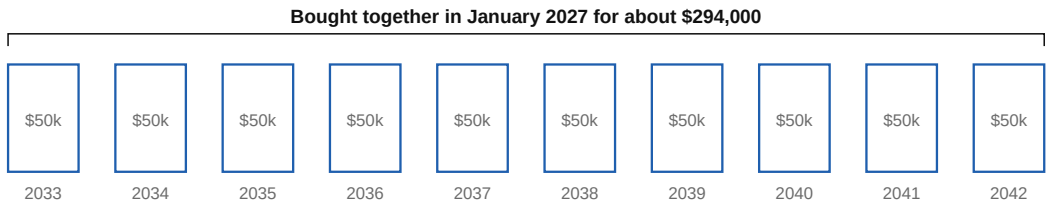


Approximate amounts at September 2026 prices. The building-floor bond costs about \$118,000 in 2028 and repays \$150,000 just before 2033. Blue marks the only money exposed to the stock market.

**If the bonds cost more than \$300,000 in January** (this happens if interest rates fall first), we buy the latest payments first and use the 2028 deposit to finish the rest before anything else. Our model says this is needed in about 1 case in 3. The payments are still fully secured once the 2028 money arrives.

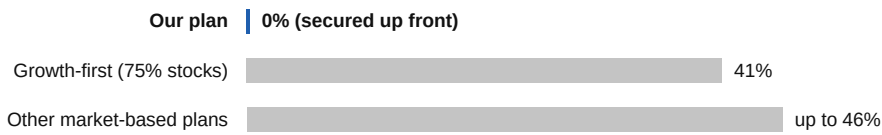
# How safe are the payments?

Each bond pays out just before its payment date. Stock-market crashes, a slow year for Laura's books, or changes in interest rates do not change what these bonds pay.



One bond for each \$50,000 payment: a "ladder".

## Chance of missing a payment if the 2028 deposit never arrives



Model estimates.

## What we are open about

| Remaining risk                                                 | How the plan handles it                                                    |
|----------------------------------------------------------------|----------------------------------------------------------------------------|
| Interest rates fall before January 2027                        | Buy the latest payments first; the 2028 deposit completes the rest.        |
| Rates fall <i>and</i> the 2028 deposit never comes             | Up to about \$31,000 of the first payment would be missing. We state this. |
| Building costs in Taiwan rise, so \$50,000 buys less each year | Payments are fixed in U.S. dollars. We name this gap rather than hide it.  |
| The U.S. government fails to pay                               | The only risk to secured payments. Remote, but named.                      |

# What Laura can tell partners in 2031

Partners who help fund a building want to know that the founder's number is real. With this plan Laura can say, truthfully, that the first \$150,000 is already owned, and that more is likely if markets hold.



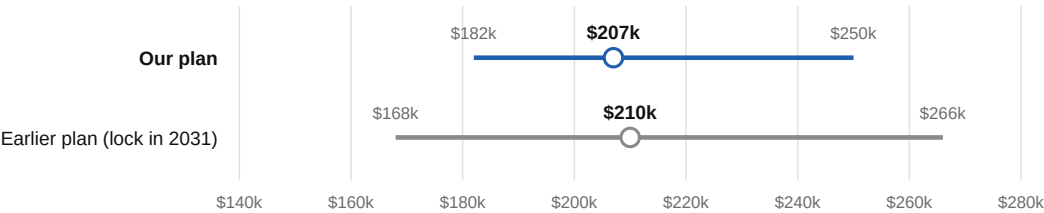
A typical case. The top of the range is fixed on the day she speaks, from what the stock fund is worth then.

|                                                |                                       |                                                 |
|------------------------------------------------|---------------------------------------|-------------------------------------------------|
| <b>\$150k</b>                                  | <b>~\$174k</b>                        | <b>~\$32k</b>                                   |
| the floor, owned three years before she speaks | typical building gift in 2033 (model) | typical money she keeps for flexibility (model) |

- The gift is the floor plus half of the stock fund, never more than the top she announced. She cannot overpromise.
- The top is reached in about 7 of 10 model cases, and in 84% of two-year periods of U.S. market history.
- In a strong market the extra stays with her, rather than raising expectations she may not meet next time.

# Results in good years and bad

## Money for the building and flexibility in 2033



Left end: a bad case (1 in 20 is worse). Circle: the typical case. Right end: a good case (1 in 20 is better). Model estimates on JPMorgan's 2026 long-term market assumptions.

## The worst six years in U.S. market history since 1928

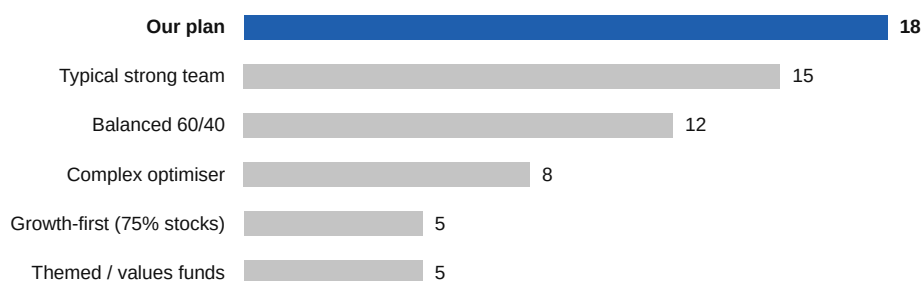


Every six-year period from 1928 to 2025, replayed as 2027-2032 and adjusted to today's assumptions.

The typical result is almost the same either way. Our plan gives up a little of the best case in return for a much better worst case, and a building floor that is fixed years before Laura speaks to partners.

## Would Laura choose this plan?

We tested the plan blind. Six one-page plans, ours and five common approaches, were stripped of names and style. Three separate reviewers, each asked to think like Laura using only the case and her own checked public words, ranked all six. Each read them in a different order.



Ranking points: 6 for first place down to 1 for last, added across three reviewers (18 is the maximum). A simulation, not Laura's real opinion.

### What the reviewers said

- The bottom of the range is money she already owns, so she cannot overpromise.
- The other plans leave her promises depending on markets until 2033.
- The plan is honest about its own costs and gaps.

**Their one criticism:** it can read as too cautious. The answer is to say clearly where the bold move is (the residency itself) and to state the price of certainty in one line.

# The competition portfolio

In the competition's trading game (\$300,000, 28 September to 6 November) we recommend holding Laura's plan as it would look in January 2028, scaled to \$300,000. Every trade then shows a real part of the strategy.



Recommended mix, about 1% in cash. Each fund must first be confirmed as available on the platform and within its holding limit; alternatives are ready.

| Holding                | Its job in Laura's plan                    | Trading note |
|------------------------|--------------------------------------------|--------------|
| Government bond funds  | Stand in for the ten payment bonds         | The promise  |
| A 2032 government note | Stands in for the \$150,000 building floor | The floor    |
| A world stock fund     | The growth money nobody depends on         | Growth       |

# How the research was organised

We treated the case as a real client file. The work ran in eight phases. Each handed a smaller, better set of material to the next, and nothing moved on without a source.



The eight phases and what each produced.

## From many questions to the few that matter



A question survived only if its answer could change a decision, it pointed to a real source, it was not trivial or already answered, and at least two of three sceptics agreed it should stay.

Why work this way? It stops a clever plan being built on an unchecked fact. Every figure in this report can be traced to an official document, a checked source, or a model we can re-run.

## Phase 1 · Foundations: what the rules and facts really say

We broke every official document into 324 numbered lines so that any claim can point to its exact sentence, checked market facts at their original sources, mapped everyone with a stake in the plan, and set trading guardrails.

### **This year's trading rules are public**

\$300,000 of virtual cash (the same as Laura's first deposit), up to 200 trades, any listed fund or U.S. government bond, no minimum number of sectors, no day trading, and notes that cannot be edited once saved. The limit on any one holding is visible only after logging in.

### **\$300,000 is only just enough**

The ten payment bonds cost about \$294,000, leaving about \$5,600 spare. Earlier in 2026 the same bonds cost more.

176 days: the ten bonds cost more than \$300,000



9 days: under \$300,000

Trading days in 2026 up to 25 September.

### **Market facts, checked at the source**

- The U.S. central bank raised its rate to 3.75-4.00% on 16 September 2026. The 10-year government bond yield was 5.17% on 25 September (an online summary claiming 4.22% was wrong).
- Taiwan: prices up 2.04% over the year to August; building costs up 6.53% over the year but about 3.5% a year since 2021.
- JPMorgan expects world stocks to return about 7.0% a year over the long run, and cash about 3.1%.

### **Surprises and traps in the case**

- Laura puts in \$450,000 and must pay out \$500,000, yet the ten payments cost only about \$294,000 today, because bonds earn interest until each payment is due.
- "A high degree of certainty" is never defined. We define it by price: the payments are bought, not forecast.
- The reserve is "set aside" in 2033, the day the first payment is due. We explain why we buy it in 2027.
- No currency is named, although the residency is in Taiwan. The rules on permitted investments apply to both deposits, so we read them as ruling out currency contracts.
- The case says she "will" add \$150,000. Testing a missing deposit is our own stress test, and we label it that way.
- In the story an advisor "decides"; the real rules say advisors may not make decisions. Every choice is the students'.

- Gains in the trading game never count toward Laura's money, and trade notes can never be edited.
- There is no approved fund list this year, so the platform may accept trades the rules ban. Older rules found online (a sector minimum, other cash amounts) are wrong.
- Wharton's sample trading note will be widely copied. Ours must carry facts only Laura's plan has.

### **People we might forget**

- The residency's staff live on fixed \$50,000 payments that buy less each year and stop after 2042.
- Taiwanese partners think in Taiwan dollars, and "dollar" is ambiguous in Taiwan, so we always write "U.S. dollars".
- Book advances come in instalments, so the 2028 deposit could be late, not only smaller. The plan's rules handle both.
- Laura's credibility is part of her income. A missed public number would hurt her twice.
- Funders dislike paying running costs. Because operations are already funded, partners' money can go to the building.
- After 2033 she may earn less while running the residency: another reason to keep some money back.

### **Two official documents were missing**

The 2026-27 Competition Guide and Infographic are now filed. The Infographic says what judges reward: "Strong teams explain the reasoning, assumptions, and tradeoffs behind their decisions." Last year about 2,300 of 6,300+ registered teams finished.

## Phase 2 · Questions: looking from every angle

Twenty question-writers, working in pairs from ten angles, produced 472 questions and collected 271 quotes.

| Angle                   | What it looked for                                                           |
|-------------------------|------------------------------------------------------------------------------|
| The case, line by line  | What the case says, leaves out, or changed from past years                   |
| Laura's own story       | Her public interviews, books and talks, and how Wharton frames her           |
| Her career and income   | How steady her income is, and what that means for the 2028 deposit           |
| Professional practice   | How pension funds and private-wealth advisers handle promises like hers      |
| Markets in 2026         | Interest rates, stocks, AI, Taiwan and currency                              |
| Future partners         | What a co-funder checks before trusting a founder's number                   |
| Judges and past winners | What strong teams do, and what a "typical strong team" will submit           |
| Her own words           | Quotes collected so they can be checked word for word                        |
| Client synthesis        | Her goals and limits in one picture; Laura as a buyer choosing between firms |
| Contrarian              | The strongest case against our own plan                                      |

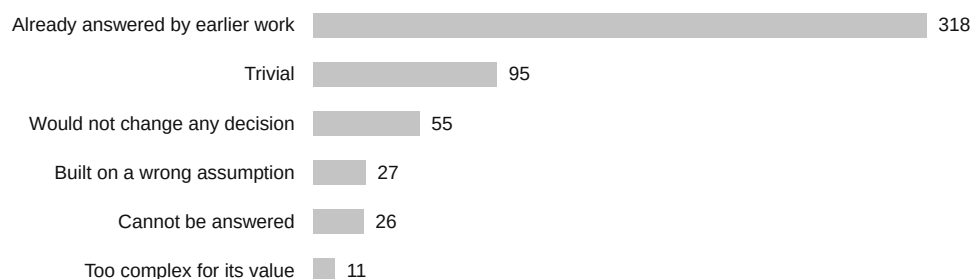
### The questions that mattered most

- How much money should stay free for flexibility after the reserve, and by what rule?
- Which "if this happens, we do that" rules must be fixed in advance?
- What do we do if interest rates fall before January 2027 and the bonds cost more than \$300,000?
- Should the trading game show Laura's first-year portfolio, or the plan once all three jobs exist?
- What does "thoughtful risk" mean for a founder like Laura?
- How should a range be told to partners without overpromising?

## Phase 3 · Filter: keeping only what changes a decision

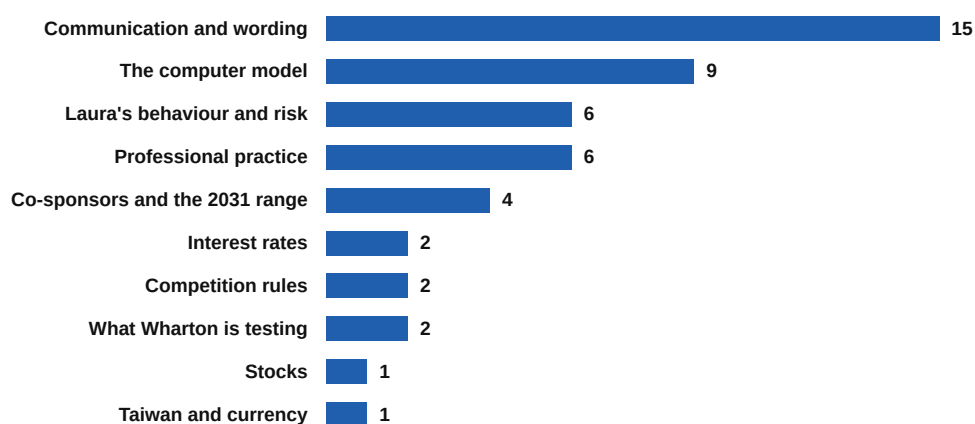
Duplicates were merged, leaving 248 questions. Each then faced three sceptics: a competition judge, a checker for questions already answered, and a statistician hunting for wrong assumptions. 48 survived.

### Why questions were cut



Reasons the sceptics gave (one question can receive several).

### Where the 48 surviving questions landed



The lesson: most questions that sound important fail a plain test. The biggest group of survivors was about how to explain the plan, not about the investments themselves.

## Phase 4 · Research: what the specialists found

Twelve specialists answered the 48 questions, and five auditors re-ran the maths and re-opened the sources. Here is everything they found, in plain words.

### Interest rates and the January 2027 purchase

- The ten bonds cost about \$294,387 at 25 September prices, \$5,613 under \$300,000. A fall in interest rates of just 0.19 percentage points would use up that cushion; the model puts the chance at about 1 in 3.
- Buy all ten as soon as the money arrives and keep any leftover in short-term government bills. Waiting for better prices or buying in stages gives no reliable gain and a 27-35% chance (model) of needing the 2028 deposit.
- If money is short, buy the latest payments first. After a one-point rate fall with no second deposit, \$31,400 would be missing instead of \$47,700. The cost: the first operating year is the one that waits.
- How big the gap could get (model): \$9,169 after a half-point fall, \$24,793 after a one-point fall, \$27,631 on 2026's worst day. A repeat of late 2008's sharp fall would need \$48,600, a third of the second deposit, so we never promise an "at most" figure.
- The 2028 deposit finishes the payments before anything else. About \$43,000 of it covers a 1.5-point fall, and \$51,000 the worst three-month fall since 1990.
- Considered and rejected: a "funding below 100%" alarm. It can never go off once every payment is bought; the right question is simply "is every payment bought?"

### Stocks and AI

- Stocks must beat about 5.2% a year, what U.S. government bonds pay for Laura's 2028-2033 dates, not zero.
- Forecasters disagree about how much extra stocks will earn: JPMorgan about 1.5 to 1.8 points a year above bonds, Vanguard about zero, all sources between -1.2 and +2.4. Stocks are there for upside, not to meet the goals.
- U.S. shares look very expensive against ten years of average profits (second only to December 1999) but normal against next year's expected profits, because profits are expected to rise 32%, led by AI chips. The plan must work either way.
- The world stock fund's ten biggest holdings (21.7%, all AI-linked) are far less concentrated than the S&P 500's (38.8%), but it still owns Asian AI chipmakers. We never claim it "halves AI exposure", and we use no Taiwan, AI or theme funds.
- Moving the growth money between 50% and 60% in stocks barely changes the typical result (-\$1,500 to +\$400). The final plan simply puts all spare money in one world stock fund.

### The computer model and stress tests

- If the 2028 deposit arrives, even at \$75,000 or a year late, the payments are never short (model). The biggest swings come from that deposit (\$75,000 instead of \$150,000 cuts the typical result by \$98,000), from fees, and from which forecaster you believe (Vanguard's view: -\$6,000).

- Fees matter: a 0.5% yearly fee on the growth money wipes out most of what stocks add. So all costs come only from the stock fund.
- The model has too few crashes (big falls in 0.3% of years, against 3.1% in reality), so we always show real history beside it. Replaying past crashes: before the floor is bought, building money falls 19-48%; after, only about 4-6%. The payments stay bought.
- Buying only 80-90% of the payments fails in 13-26% of futures if the second deposit never comes. Buying all ten costs just \$900 to \$5,200 of typical building money. Full certainty is cheap.
- The "safe floor in 2028, rest in stocks" design beat locking growth in 2031 on bad cases, at the same stock risk, so it became the final plan.
- Considered and rejected: locking 80-90% in 2031 and giving "90%, capped" (it announced about \$17,000 less than was owned); a model-based "80% confident" range (right only 62-93% of the time); locking the floor in three steps (a \$2,000-6,000 gain, not worth a second rule).

### **Taiwan, currency and building costs**

- Building costs in Taiwan are expected to rise about 2-3.5% a year, so each U.S. dollar buys roughly 12-20% less building by 2033. 2026's +6.53% jump is a spike to mention, not the trend.
- "Certain" means certain in U.S. dollars. At 2.5% inflation, \$50,000 is worth \$42,063 in 2033 and \$33,681 in 2042 in today's money. We name this gap once.
- Any Taiwan-dollar figure is an illustration only (NT\$31.82 per U.S. dollar on 18 September 2026). By 2031 the exchange rate could move about 11-13% either way.
- No currency conversion or hedge before the building decision: a currency contract is a banned derivative, and it would cost about 5.9% now.
- Cushions: the Taiwan dollar tends to weaken when U.S. stocks fall, which helps; the 1995-96 missile crisis moved it by at most 4.4%. War is outside all the data. It would threaten the project, not the bought payments.

### **Who Laura is**

- She holds two attitudes to risk: leaps whose cost she carries herself are fine, but letting down the people who backed her is worse. So money others rely on takes no risk. (Her public phrases are background for the team only, not text for the IPS or notes.)
- Bold in her career does not mean bold with investments. We never use her career story as a reason to own stocks, and since she has made no public statement about investing, we never describe her "investment philosophy".
- She has taught financial literacy to teenagers and interned in trading and in consumer protection. Expect plain words and named costs: she dislikes a show-off finance tone, not finance.
- Her public projects put access first and grew once the community backed them. "Operations first, partners invited later" fits her.
- She would welcome: purpose first, costs and gaps named, discarded drafts shown, rounded numbers, and what is secured ("10 of 10 years funded") before any dollar amounts.
- She would reject on sight: the case's pull quote presented as hers; puns on her book titles; her identity or Taiwan as a reason for a holding; false precision; hidden costs; one-sided risk stories; wrong facts about her work.
- Dropped: "floor first, then leap" as a trait of hers. It came from a reporter's sentence, not her own words.

## Co-sponsors and the 2031 range

- Funders count only unconditional pledges and cash. So Laura leads with what is bought: in U.S. dollars, not forecast, barring a U.S. government default. Never "guaranteed": she only "expects" to give more.
- Confidence both ways: below the bottom only if the U.S. government defaults; the top is reached if stocks are no lower two years later (true in 84% of U.S. two-year periods since 1928; 73% in the model).
- Words: "owned", "expected if stocks hold their value", "kept". Never "promised". Strong markets add to the money she keeps, not to the gift, though she may still choose to give more.
- Flexibility needs a rule, not leftovers. Keeping half the fund gives about \$32,000 in the typical case, roughly three to four years of Taiwan building-cost rises.
- Considered and rejected: a model-based top with no cap; converting to Taiwan dollars early; a "10% of wealth" rule of thumb; giving three-quarters or all of the fund.

## Behaviour and risk, goal by goal

- Judge risk one goal at a time, not as "moderate" overall. The payments need no risk and can bear none. The stock fund can bear losses: her living costs are covered elsewhere and nothing is withdrawn before 2033.
- Her ability to take risk is lowest before the 2028 deposit, which depends on her career, and the plan holds no stocks then.
- The payments need only about 1.05% a year to be met. More stocks would add about \$11,000-14,000 in the typical case but cost \$41,000-56,000 in a bad case (model).
- The stock fund falls in about a quarter of years (model). In the worst tenth of 2031-32 outcomes, the typical gift is still about \$169,000, and never below the floor.
- Decide in advance: never sell stocks to protect a gain or to "wait for a recovery". Rules change only when her circumstances change, never because markets move.
- Her big bets are already her career and books, so the portfolio holds only broad funds, and the money she keeps sits in short-term government bills.

## What the judges and the case designers test

- The three written deliverables are judged together, the notes are quoted exactly, and the game's ranking counts for little.
- This year's case is about certainty and credibility: "certainty" appears 4 times, "credible" or "credibility" 3 times, "growth" once. None of the five "must" tests mentions growth.
- Creativity means fit to Laura, not clever investments. Rivals using AI may reach the same design, so our edge is Laura-specific reasons and real evidence of the team's own process.
- Simplicity wins: a past semifinal judge praised "simple, elegant ideas", and the 2022-23 champions said "less is more".
- "Supported, tested or refined" is a menu, not a quota. A note may say "refined" only if the team's log shows the change was decided before the trade.
- The strategy freezes on 6 November, so every rule must be fixed by then. Changes based on research before that date are fine.

- Breaking the format rules means exclusion, so the IPS stays well inside its limits. The working checklists hold the details.

### **What professional investors do**

- Pension funds "lock down the benefits promised, then grow surplus assets". Our design is standard professional practice.
- Professional policy statements fix rules for market turmoil in advance, and can even state a policy of not rebalancing. Ours never rebalances.
- The competition points to a professional code: know the client, follow the stated rules, disclose fees plainly, keep records.
- A typical adviser charges about 1% a year. Charging costs only to the stock fund costs about \$3,000 (typical); charging them on everything, about \$31,000 (model).
- Returns are stated as a range from two published forecasters, never as a target.
- Considered and rejected: inflation-linked U.S. bonds (they track U.S. prices, not Taiwan's), and buying only part of the payments.

### **How to talk about the plan**

- Central idea: a split of jobs. Money others rely on is bought when it arrives; money nobody relies on is fully in world stocks.
- Pitch: purpose first; each deposit buys a promise; she tells partners only what she owns. No numbers, and never "bought in January 2027", which fails in about 1 case in 3.
- One word per job: "certainty" for the payments, "confident" for the range, "credibility" for her standing. Avoid "guaranteed", "risk-free", "100%", jargon, return targets and "innovative".
- Name the two gaps plainly (the January price, and what fixed dollars buy in Taiwan) and keep risk talk to about 15% of the words.
- In the trading game, funds never mature, so say "moves like" or "stands in for", never "matched" or "locked". Measured against fixed payments, cash is actually the riskier choice: a 0.1-point rate move changes the payments' value by about \$2,900.
- Name three trade-offs as "gives up X to get Y", and test the text on six outside readers, three with a finance background and three without.

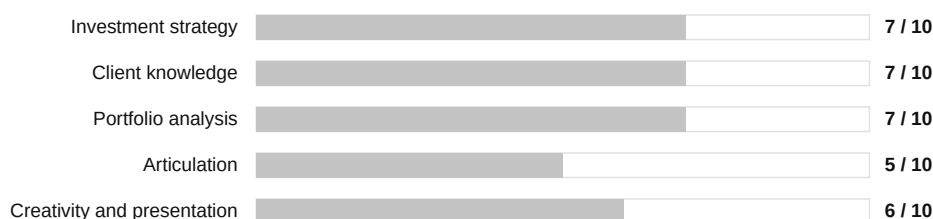
### **What the auditors corrected**

- The chance that the bonds cost more than \$300,000 is about 1 in 3, not 1 in 4.
- "At most" promises and a "0.01% chance" were dropped: the model's 1-in-10,000 event actually happened in 2008.
- About \$26,000, not \$25,000, covers a one-point rate fall.
- The back-up fund for the building floor repays no fixed amount, so we never say it "repays \$150,000".
- Building costs are "expected", not "certain", to rise; one +8.1% figure was construction wages only.
- JPMorgan's bond assumption is 0.5 to 1 point out of date, so results are shown with Vanguard's view too.

## Phase 5 · Stress tests: four teams tried to break the plan

Four teams read the draft plan and tried to break it. Everything they found was either fixed or answered in the final plan.

### A Wharton judge



The judge's scores for the earlier draft: 32 of 50. With ten fixes, about 38 of 50. The Top 50 is roughly the top 2% of finishing teams.

- Too dense: about 55 ideas in 550 words. Fixed: about 12 ideas in 10 short blocks, on four dates.
- The pitch opened with a ban on stocks that the trading game breaks on day one, and it skipped the residency. Fixed: purpose first, and "each dollar has one job".
- A bond expert's red pen: short bonds used for dated money, no reason given for buying the latest payments first, and a withdrawn number. All fixed.
- Six named risks read as defensive. Fixed: two gaps plus a U.S. default, named once.
- The best story was unused: "we tested a 75%-stock plan and dropped it" (it missed payments in about 3% of model futures, and about 40% without the 2028 deposit). It is now one of the trading notes.

### "Laura" (a simulation grounded in the case and her checked public words)

- Would likely choose us: once bought, the payments stop depending on markets or her future earnings; the 2031 figure never needs taking back; gaps are named; she keeps her own decisions; no identity-based holdings.
- Pushed back on: codes and bans before purpose, "promised" for money still at risk, unstated fees, and personality labels. All fixed.
- Found it timid. Fixed by explaining (the payments cost about 98% of her first deposit), not by adding stock, and by saying what stocks are for: a bigger gift and flexibility in good markets. A fall costs only the extra.
- The questions she would ask: which year waits if 2027 prices are high (2033); what she pays; what stays hers; whether a crash changes the plan (no).

### A rival team's best strategist

- The attack: "Buy a bond in January 2028 that repays the full \$150,000 by 2033, put the rest in one world stock fund, and trade nothing in 2031." We adopted it.

- Why it is better: the lowest number she announces is fixed three years early (\$150,000, against \$134,000 in a bad case, or \$93,000 after a 1929-style crash, for the old design); the worst six-year stretch leaves \$169,000 against \$120,000; the typical result is the same.
- What we accept in return: a typical gift of \$174,000 instead of about \$186,000-188,000; a best case \$16,000 lower; strong markets grow her kept money rather than the gift.
- Where the old design still wins: if a crash recovers before 2031, as in 1973-78.

#### **Pre-mortem: imagine we failed. Why?**

- The likeliest failure is ownership, presentation and process, not the strategy.
- The team does not own the plan: each student writes their own reasons, without AI, before every vote; devil's-advocate pairs; two-minute explain-backs.
- A format breach, which is fatal: strict limits, finish early, check the final PDF.
- Too dense: about 12 ideas, tested on six outside readers.
- AI wording in permanent notes: draft offline, check for copied phrases, keep a dated log of AI use.
- Process: roles agreed now, deadlines two days early, weekly check-ins. School holidays and exams (13 October to 5 November) may clash.
- Cheap but fatal: submit a day early (22 October, 5 November); roster in by 7 October; no contact with the client; no paid help; all six members involved.

## Phase 6 · Decision: the ten insights behind the plan

| Question                                                           | What we concluded                                                                                                                                                                                                                                         |
|--------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| When two designs give the same typical result, which wins?         | The one with the better bad case and fewer rules: \$182k vs \$168k in a bad case, \$207k vs \$210k typically (model); \$169k vs \$120k in the worst historical stretch.                                                                                   |
| What job is Laura hiring a firm for?                               | To make promises no market or future contract can take back: payments first, then a building floor from her own earnings.                                                                                                                                 |
| Is "high certainty" true?                                          | Yes, because it is bought, not forecast, with two honest gaps: the January 2027 price can exceed \$300,000 (about 1 case in 3), and fixed dollars buy less as prices rise (at 2.5% inflation, the 2042 payment is worth about \$33,700 in today's money). |
| Why so little stock for a founder who likes bold moves?            | The caution is forced by arithmetic, not chosen for her personality: the payments use about 98% of the first deposit. All money nobody relies on is fully in stocks.                                                                                      |
| How is the 2031 range set without overpromising?                   | By a rule: bottom = what she owns; top = a market value on the day, capped; any extra stays hers; the chance of reaching the top is stated.                                                                                                               |
| What makes this more than a textbook plan?                         | Each deposit buys one promise when it arrives. The second deposit's own \$150,000 becomes the building floor. That rule is specific to Laura.                                                                                                             |
| What should the trading game show?                                 | Her plan on the first day all three jobs exist (January 2028), scaled to \$300,000, and every note says so.                                                                                                                                               |
| Which words survive a permanent note?                              | Never "match", "promised" or "guaranteed". Say "moves like" or "stands in for": most bonds in the long bond fund mature after 2042.                                                                                                                       |
| How do the three notes show "supported, tested, refined" honestly? | One is tested against numbers written down before the trade; one is "refined" only if the team's log shows the change was decided first; one tells the true story of dropping growth-first after testing.                                                 |
| How much fits in a 500-word IPS?                                   | About 12 ideas in 10 short blocks, one number, then tested on six outside readers.                                                                                                                                                                        |

## Phase 6 · Decision: what changed from the earlier plan

| Change                                                                          | Why                                                                                                                  |
|---------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <b>Vote before any trade</b>                                                    | Notes are permanent. Trading before the team decides could contradict the plan later.                                |
| <b>Central idea: a split of jobs</b> , not a ban on stocks until 2028           | A ban is broken on day one by the stock fund in the trading game.                                                    |
| <b>Game portfolio: three jobs</b> instead of roughly 65% bonds and 35% stocks   | Three jobs give three distinct, honest notes. Four trades cost \$100 in fees.                                        |
| <b>Notes written by students</b> , short, checked by two outside readers        | Notes cannot be edited, and the AI policy requires the students' own words.                                          |
| <b>Building floor bought in 2028</b> , not 80% of a stock sleeve locked in 2031 | Same typical result, better bad cases, fewer rules. Cost: a smaller typical gift (\$174k vs about \$186-188k).       |
| <b>A capped 2031 range</b> with the chance of reaching the top                  | A top taken from a model landed in the range only 62-93% of the time.                                                |
| <b>Certainty defined by price, with two named gaps</b>                          | "Fits inside \$300,000" and "bought in January 2027" fail in about 1 case in 3, so they cannot be headlines.         |
| <b>Buy on arrival, latest payments first</b>                                    | If short of money, this leaves less unfunded (\$31.4k vs \$47.7k if rates fall 1 point and the deposit never comes). |
| <b>2028: finish the payments first</b>                                          | The joint worst case is named once: \$11,957 or \$31,413 of the first payment unfunded if rates fall 0.5 or 1 point. |
| <b>2033: no currency conversion or hedge</b>                                    | Currency contracts are derivatives, banned in the game; a hedge would cost about 5.9%.                               |
| <b>Costs paid from the stock fund only</b>                                      | A 1% fee on the stock fund costs about \$3k (typical); on everything, about \$31k.                                   |
| <b>A simple IPS</b> : 10 blocks, about 12 ideas, one number                     | An early draft had about 55 ideas; last year's lesson was "over-complex".                                            |

### What we considered and rejected

- Waiting for better bond prices or buying in stages: no reliable gain.
- Inflation-linked U.S. bonds: they track U.S. prices, not Taiwan building costs.
- A Taiwan fund, AI or theme tilts: concentration and tokenism.
- Locking growth in 2031 (the earlier design): kept as the documented alternative for the team's vote.

Kept from the team's original thinking: buy all ten payments first, latest first, and hold them to the end.

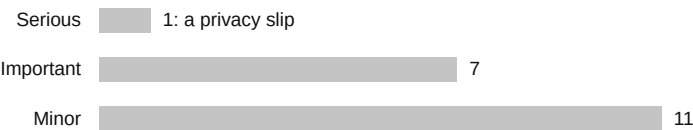
# Phases 7 and 8 · Blind test and final check

## Phase 7 · Blind test

Six anonymous plans were written to the same length and layout, and a fairness checker removed loaded words. Three reviewers playing Laura ranked all six; ours was first every time (full results on the "Would Laura choose this plan?" page). Their shared request, to show the balance between caution and ambition more clearly, is now on the team's list.

## Phase 8 · Final check

A final checker compared every output with the plan and re-ran the numbers.



Problems found. 18 of 19 were fixed; one minor item was correct as an illustration and left.

- **Serious:** one filed question quoted a line we had excluded for privacy. The text was removed.
- **Important:** outdated figures that had since been corrected; one of Laura's own words placed where only the case's words belong; a claim that the plan "never misses" without its stated exception; and funds named without the "check availability first" warning.
- After the fixes, every figure matches the final plan, and all files agree on dates, portfolio, notes and rules.

## Decisions for the team

The team decides every one of these. Our recommendation is shown for each.

| Decision                                             | Our recommendation                                                                                                                                                        |
|------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Buy all ten payments first?                          | Yes. Each student writes why, in their own words, before the vote, and the vote is recorded.                                                                              |
| Floor bought in 2028, or growth locked in 2031?      | 2028. Choose the design all six students can explain in two minutes.                                                                                                      |
| What the trading game shows                          | Laura's plan in January 2028, scaled to \$300,000. If outside readers think her real 2027 money holds stocks, show her first-year portfolio instead.                      |
| Share of the stock fund given to the building        | Half. The half she keeps is about three years of Taiwan building-cost rises.                                                                                              |
| More stock, since reviewers found the plan cautious? | No, by default. If wanted, a smaller floor (about \$100,000-134,000) is the option. Either way, state what certainty costs.                                               |
| Labels and roles                                     | The team's own labels, identical everywhere, before the first note. Every student owns a note or reflection and a part of the IPS.                                        |
| Keep the repository public until 4 December?         | The team leader decides by 2 October, weighing the risk of copying.                                                                                                       |
| Checks on the trading platform                       | Record the holding limit, confirm each fund is listed, test the note box without saving, and check for any minimum-activity rule. Never add trades just to reach a count. |

### Timeline

| When                | Action                                                                                                                            |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Before trading      | Record the three votes. Log in to the platform: record the holding limit, check each fund is listed, test the note length safely. |
| By 2 October (U.S.) | Place the trades. Each note is written by one student and checked by another.                                                     |
| 23 October          | Submit the Trading Notes Analysis.                                                                                                |
| 6 November          | Submit the Investment Policy Statement. Trading ends and the strategy is final.                                                   |

# Lessons and plain-word glossary

## Three lessons from this work

- 1. **Match money to its job.** Money someone relies on is protected; only spare money takes risk.
- 2. **Check before you build.** Several popular "facts" (a quote, a story about Laura, a rule, a market figure) were wrong.
- 3. **Simple beats clever.** The recommended plan has two purchase dates and nothing to manage afterwards.

## Glossary

|                                  |                                                                                             |
|----------------------------------|---------------------------------------------------------------------------------------------|
| <b>Bond</b>                      | A loan to the U.S. government that repays a fixed amount on a fixed date.                   |
| <b>Ladder</b>                    | Several bonds maturing in different years, one for each payment.                            |
| <b>Building floor</b>            | The \$150,000 for the building that Laura owns from 2028, whatever markets do.              |
| <b>Stock fund</b>                | One fund that owns thousands of companies around the world.                                 |
| <b>Interest rates fall</b>       | Bonds become more expensive to buy; this is why the January price matters.                  |
| <b>Model</b>                     | A computer simulation of thousands of possible futures. It gives estimates, not forecasts.  |
| <b>Bad / typical / good case</b> | The result that 1 in 20 futures is worse than / the middle result / 1 in 20 is better than. |
| <b>Red team</b>                  | A group asked to attack a plan and find its weaknesses.                                     |
| <b>Trading game (WInS)</b>       | The competition's simulator, with \$300,000 of virtual cash.                                |
| <b>IPS</b>                       | Investment Policy Statement: the rules the portfolio will follow.                           |

Where the detail is: the Trading Notes pack and IPS specification are the working checklists; the 831-page research record holds every question, source, model and check behind this report.